

Are you responsible?

Enhancing global explainability and reducing explanation latency in phishing email detection

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Abstract. Machine learning models for phishing email detection can achieve high accuracy. However, they often sacrifice explainability for predictive performance and are therefore treated as black boxes. Existing explainable AI (XAI) techniques typically focus on explaining individual predictions, but they often lack global interpretability and can be computationally expensive. This study aims to improve the global interpretability of phishing detection models while reducing the cost of generating local explanations. In contrast, global surrogate models aim to approximate the decision behaviour of the original model across the entire dataset, making them more suitable for understanding general model logic and failure patterns. This study addresses the lack of global interpretability in phishing detection models while reducing the computational cost of generating local explanations. To achieve this, we propose to use global surrogate models to approximate and explain the behavior of the original models. In particular, we adopt Explainable Boosting Machine (EBM) and GAMI-Net as global surrogate models for Random Forest and Deep Neural Network teacher models. We evaluate the surrogate models using accuracy fidelity, F1 fidelity, and a proposed Error Fidelity metric, which measures the ability of the surrogate to explain the original model’s unexpected behavior. Experimental results show that both EBM and GAMI-Net achieve high fidelity for both teacher models.

1 Introduction

As the digital world is evolving, cyber-crime especially email phishing is a threat to the internet users. Current techniques to detect phishing are primarily based on machine learning models that are reliable in terms of accuracy, however, it’s trustworthy is not definite. Sarker et al. (2024) stated that the transparency of the model contributes to its worthiness and accountability. Phillips et al. (2020) also reinforce the importance as an intelligence needs to be explainable, meaningful, reflect on its output and acknowledge it’s limitation. Hence a global explanation of a model can gain trust from businesses, customers, help diagnose incidents, and acknowledge its weak points. This assists the models to aligns with the principles identified in Phillips et al. (2020).

However, current techniques of explainable AI study on email phishing detection models are only interpret for single email. This helps the model responsible for its own output, but it does not directly help the model meaningful overall. Moreover, the latency can be a pain point since current techniques requires expensive operations.

Therefore, our we aims to uncover the black-box nature of the models then give a global explanation and reduce the latency to compute explanation for individual email.

2 Related Work

2.1 XAI methods to explain email phishing detection models

Traditional machine learning algorithms has been widely used to detect phishing emails. Safi and Singh (2023) analysed that different machine learning algorithms dominants in email phishing detection methods. Algorithms such as Random Forest, Support Vector Machine or Multi-layer Perceptron are identified to be studied in vast number of studies. However, Radulovic et al. (2021) argues that machine learning model such as Random Forest and Neural Networks are a black-box classifiers. Current techniques for explaining phishing email detection model shown in Sharma et al. (2025) shown that LIME and SHAP are two most common methods to explain a black box model in cybersecurity, specifically email phishing. However, these methods are mainly classified as local post-hoc methods which only produce interpretation per sample instead of explain an overall behavior.

2.2 Global Post-hoc Model Agnostics Methods

There are several global post-hoc model agnostics methods that we can adopt to explain email phishing models. Friedman and Popescu (2008) used Rule Ensembles to build an intrinsically explainable models using if-then rules. However, if the complexity grows, explainability can be reduced and it is instable. Apley and Zhu (2020) on the other hand used Accumulated Local Effects which accumulate small changes to observe changes in the predictions. However, the authors also state that it may be instable and does not scale well. Bastani et al. (2018) used a decision tree as a global surrogate models to interpret the original models. However, the author also, pointed out the simple models such as decision tree may not fit well on the training data with low-depth tree.

2.3 Glass-box models

To address the under-fitting problem of simple global surrogate models. Alternative choices for global surrogate models can be adopted. Generalized Additive Models (GAM) is a family of additive models that can be interpretable and visualized (Wood, 2017).

3 Method

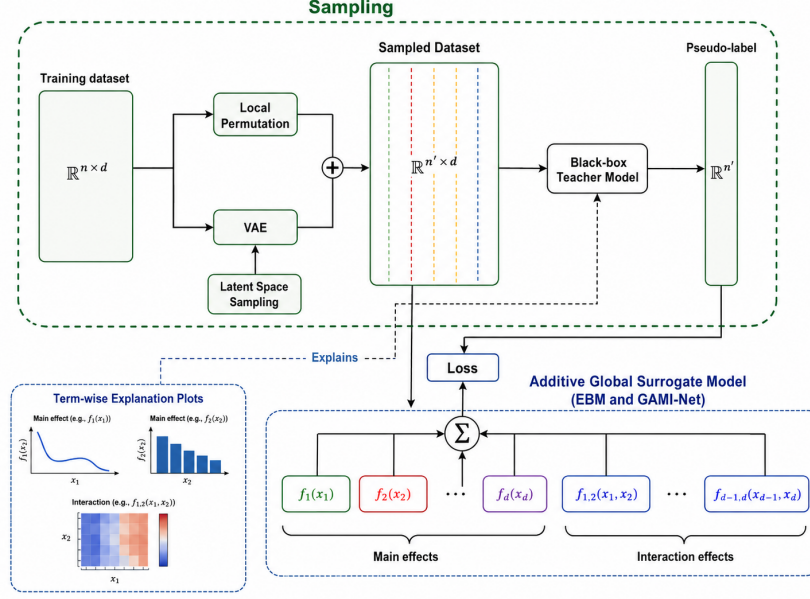


Fig. 1: Overview of the methodology

3.1 Problem formulation

We study global post-hoc, model-agnostic explanation for phishing email detection. Let the original labelled dataset be $D = \{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathbb{R}^d$ denotes the processed feature vector of an email and $y_i \in \{0, 1\}$ denotes the ground-truth class label, with $y_i = 1$ indicating phishing and $y_i = 0$ indicating legitimate email. Let P_X denote the input distribution over the feature space $\mathcal{X} \subseteq \mathbb{R}^d$.

A black-box teacher model F is first trained for phishing detection. Rather than directly explaining its internal parameters, our aim is to learn an interpretable global surrogate model S whose behaviour approximates that of the teacher over the input distribution. Formally, the goal is to obtain a surrogate such that

$$S(x) \approx F(x), \quad \forall x \sim P_X.$$

3.2 Global additive surrogate models

To approximate the behavior of the black box model, we adopt a global surrogate model that is intrinsically interpretable and behaves approximately similar to the

original models. We define original black-box model as $F(x)$ and global surrogate model as $S(x)$. If two models behave similarly, their prediction should align with each other. Therefore, our learning objective is:

$$S^* = \arg \min_{S \in \mathcal{S}} \mathcal{L}(S; D')$$

Where D' is the training dataset for $S(x)$ defined as $D' = \{(x_i, F(x_i))\}_{i=1}^{|D'|}$. Hence, the goal is to minimize the loss function:

$$\mathcal{L}(S; D') = \frac{1}{|D'|} \sum_{x_i \in D'} - \left[F(x_i) \log S(x_i) + (1 - F(x_i)) \log (1 - S(x_i)) \right]$$

For the choice of Global Surrogate Model, we adopt a family of generalized additive models (Hastie & Tibshirani, 1990). These models include Explainable Boosting Machine (Lou et al., 2013) and GAMI-NET (Yang et al., 2021). A generalized additive model is an additive model where each term only predicts the output based on one feature only for main effects. Therefore, the contribution for each feature can be plotted on a 2D plot. Another optional term that can be added is the interaction terms where terms predict using 2 features. Hence, it can be a 3D plots or a heatmap. The definition for the Global additive surrogate model is:

$$\text{logit}(S(x)) = \beta_0 + \sum_k f_k(x_k) + \sum_{i < j} f_{ij}(x_i, x_j)$$

Where $f_k(x_k)$ is a term that learn on 1 feature only and $f_{ij}(x_i, x_j)$ is a term learn on the interaction terms of 2 features. For EBM, each term is a decision tree while for GAMI-Net is a sub-neural-network.

3.3 Sampling Method

Due to the curse of dimensionality, the space for sampling grow exponentially with number of dimensionalities. However, the necessary data samples for training surrogate model are lying near the decision boundary of the original model. Because original model are trained on the original training dataset, model's decision structure is influenced by the training data distribution, so sampling near the data distribution may represent the decision behaviour of the original model. Hence, we adopt 2 method of sampling includes local perturbations and Variational Auto Encoder (VAE) sampling.

Local perturbation This method of sampling will randomly select data samples from training dataset and apply gaussian noise scaled by the standard deviation of a cloud of k nearest neighbor samples to generate new synthetic data samples.

$$\tilde{x} = x + \lambda \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_k^2)$$

VAE sampling This method of sampling will train a VAE on the original dataset to learn the underlying data distribution in the latent space. New synthetic data will be generated by sampling in the latent space of VAE. This enable broader coverage of data distribution beyond local perturbations.

$$z \sim \mathcal{N}(0, I), \quad \tilde{x} \sim p_{\theta}(x | z)$$

3.4 Dataset construction

The dataset was .by integrating multiple publicly available phishing and legitimate email corpora into a unified business-email dataset for phishing detection model training. These sources include Nazario Phishing Corpus (Nazario, 2007), Kaggle phishing datasets (Kaggle, 2024), and the Enron Email Dataset (Cohen, 2015). In total, 132,289 samples are collected.

3.5 Data analysis and preparation

The dataset was analysed to examine class composition, attack-type coverage, missing values, and key behavioural indicators relevant to phishing detection. After construction, the records were cleaned, deduplicated, and standardised into a unified schema containing message content, sender metadata, URLs, attachments, labels, and derived indicators such as link presence, urgent language, reply-to mismatch, and suspicious sender patterns. Descriptive analysis was then conducted on label distribution, attack types, missingness, body length, link frequency, and urgent wording to assess dataset quality, potential class imbalance, and phishing-behaviour coverage. Before modelling, additional lexical, structural, sender-domain, URL, and attachment features were extracted, and the dataset was split into training, validation, and test sets using a 70:10:20 ratio. Numerical variables were imputed and scaled, while categorical variables were encoded, resulting in a processed feature space of 292 variables for downstream modelling and interpretability analysis.

Label Distribution in Raw Business Phishing Dataset

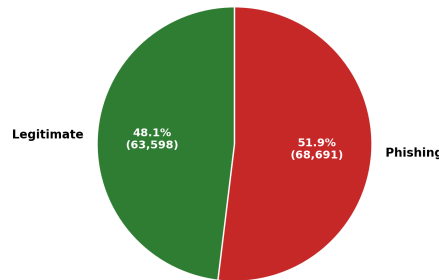


Fig. 2: Label distribution

Class distribution in figure is relatively equal with 51.9% is phishing emails. This will prevent the model to be bias to one class.

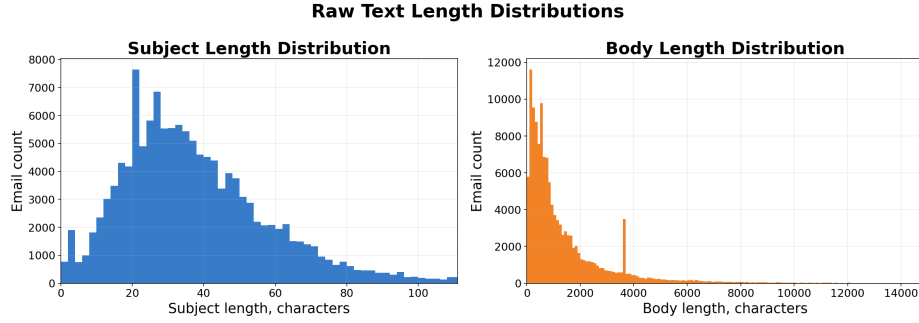


Fig. 3: Subject and body length distribution

From figure 2, it can be observed that the subject length of subject length is slightly right skewed and body length is heavily right skewed. This may be because the subject length may be restricted of character and often short while body length are often short for quick emails sometimes varies up to thousand of characters

attribute	mean	median	std	min	max
label	0.52	1.0	0.50	0.0	1.0
num_links	1.65	0.0	10.29	0.0	3133.0
has_links	0.47	0.0	0.50	0.0	1.0
num_attachments	0.00	0.0	0.00	0.0	0.0
has_attachment	0.00	0.0	0.00	0.0	0.0
has_urgent_words	0.09	0.0	0.29	0.0	1.0
sender_replyto_mismatch	0.00	0.0	0.00	0.0	1.0
suspicious_sender_domain	0.04	0.0	0.19	0.0	1.0
suspicious_attachment_type	0.00	0.0	0.00	0.0	0.0
subject_length	38.12	34.0	31.90	0.0	7169.0
body_length	1857.21	790.0	12460.84	1.0	1829070.0

Table 1: Summary statistics of dataset attributes

4 Experimental Setup

4.1 Data Preparation

First the emails text field are extracted meaningful features by feature engineering into fields such as “Urgent word counts”, “Length”. Then missing values

are imputed which numerical data is imputed by median and normalize with standardization while numerical data is imputed with most frequent value. For high-cardinality categories, they are label encoded while low cardinality categories are one-hot encoded. After the data processing, the dataset results in 292 features. With high dimension, it contributes to the low interpretability of the models.

4.2 Blackbox teacher setup

In this experiment, two teacher models are considered. The first model is Random Forest (RF) and the second model is Deep Neural Network (DNN). Random Forest teacher model is implemented with 300 trees with random state 42 for reproducibility. Because Random Forest implemented in this experiment use 300 trees and the prediction output is a collective decision of all trees, global and local understanding is hard to achieve. Deep Neural Network in this experiment uses 2 hidden layers 128 and 64 neurons with ReLu activation function. This implies the model has 45825 learnable parameters. Although the model is extremely deep, with the non-linearity and complex feature interactions, the model behaves as a black-box hence global and local understanding is hard to achieve.

4.3 Sampling Setup

Local perturbation sampling applied to real training samples for the teacher models. For each sample, the k-nearest-neighbor cloud was set at $k = 25$. Gaussian noise is added with using the standard deviation with scale 0.15. Then, for one-hot and label encoded features, the value is floor down and ground to nearest whole value within the range of training samples. As the result, 300 000 instances are sampled. Variational Auto Encoder use in this experiment configured with latent space of 20 dimensions and hidden dimensions of 256, KL-divergence weight $\beta = 1.0$, learning rate of 0.001. For sampling, the temperature is set at 0.85. Similar to local perturbation method. Processed categorical data is floor and grounded accordingly.

4.4 Global Surrogate model setup

EBM setup in this experiment uses interactions = 5, validation size=0.15, outer bags=8, inner bags=0, learning rate=0.03, max rounds=5000, early stopping rounds=100, max bins=256, max interaction bins=64, n jobs=-1, and random state=42. This setup allows EBM to achieve higher fidelity with interaction term while maintaining interpretability.

GAMI-Net is configured as interact num=5, subnet arch=[40,40,40,40,40], interact arch=[40,40,40,40,40], batch size=512, main effect epochs=50, interaction epochs=50, tuning epochs=20, backpropagation learning rate =[0.0001 , 0.0001 , 0.0001], early stop threshold=[30,30,20], heredity=True, reg clarity=0.1, validation ratio=0.15, and random state=42. For interpretation, top 10 important features of both EBM and GAMI-NET is plotted with top 5 interaction term.

4.5 Evaluation Metrics

Primary Evaluation Metrics Let the prediction from the global surrogate model and the original model on a sample defined as:

$$\hat{y}_i^F = \mathbf{1}[F(x_i) \geq \tau], \quad \hat{y}_i^S = \mathbf{1}[S(x_i) \geq \tau].$$

where τ is the confidence threshold

Accuracy Fidelity Accuracy Fidelity measures how accurately the prediction produce by the surrogate model on original model predictions (Andrews et al., 1995). A higher accuracy fidelity indicates stronger behaviours agreement between surrogate and original model.

$$\text{Fidelity}_{Acc}(S, F) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i^S = \hat{y}_i^F].$$

where S is the surrogate model and F is the original model.

F1 Fidelity Similar to Accuracy Fidelity also measure how accurate if the alignment of the prediction of surrogate model and original mode while accounting for both precision and recall across the predicted classes (Andrews et al., 1995).

$$\text{Fidelity}_{F1}(S, F) = F_1(\{\hat{y}_i^F\}_{i=1}^N, \{\hat{y}_i^S\}_{i=1}^N).$$

Proposed supplemental metric – Error Fidelity Although accuracy and F1 fidelity is strong in measuring how faithful the surrogate model compares with the original model, it may be weak in measuring the capability of explaining teacher unexpected behaviour. Consider the following example:

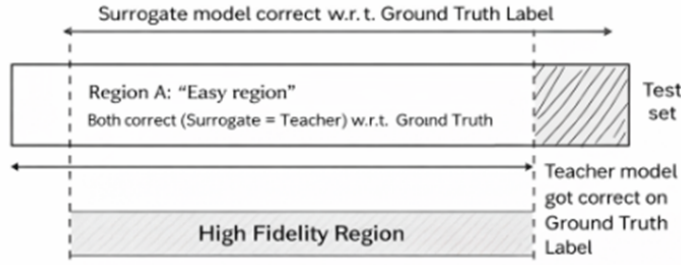


Fig. 4: Surrogate model achieves high fidelity but does not agree with the teacher model on teacher unexpected behavior.

Although the surrogate achieved high fidelity with original model, it failed to agree with original model on teacher errors on the ground truth label. According

to Phillips et al. (2020), to gain social acceptance, if the output is unexpected from the system, the explanation may need to assist users to understand the reason. Therefore, this experiment proposes Error fidelity which measures the accuracy fidelity given that the output of original model is mismatched with ground truth.

Let the error set be defined as:

$$\mathcal{E}_F = \{x_i \mid \hat{y}_i^F \neq y_i\}.$$

Then the error fidelity is defined as:

$$\text{ErrFid}(S, F) = \frac{1}{|\mathcal{E}_F|} \sum_{x_i \in \mathcal{E}_F} \mathbf{1} [\hat{y}_i^S = \hat{y}_i^F].$$

5 Result

5.1 Teacher performance on test set and fidelity on each other

Teacher	Accuracy	F1	Peer Fidelity	Peer Error Fidelity
Random Forest	0.9433	0.9450	0.9381	0.4694
Deep Neural Net	0.9149	0.9182	0.9381	0.7047

Table 2: Performance of teacher models on the test set and peer fidelity.

From table 2, despite both teachers does not learn from each, their peer fidelity is high at 0.9381. However, each model their peer error fidelity is relatively low.

5.2 Accuracy and F1 fidelity of each surrogate to its teacher

Teacher	Surrogate	Accuracy	F1
Random Forest	EBM	0.9260	0.9280
Random Forest	GAMI-Net	0.8934	0.8936
Random Forest	Decision Tree Depth 5(Base Line)	0.8311	0.8287
Random Forest	Logistic (Base Line)	<u>0.8013</u>	<u>0.8093</u>
Deep Neural Net	EBM	0.9402	0.9425
Deep Neural Net	GAMI-Net	0.9137	0.9142
Deep Neural Net	Decision Tree Depth 5(Base Line)	0.8491	0.8553
Deep Neural Net	Logistic (Base Line)	<u>0.8093</u>	<u>0.8093</u>

Table 3: Accuracy and F1 fidelity of each surrogate model with respect to its teacher model.

From table 2, it can be observed that EBM and GAMI-Net achieved high fidelity with both Random forest and Deep Neural Net. Comparing with baseline of Logistic regression, both models shows significant dominance, suggesting that both models picked up more complex behaviour of original models than just a linear relationship. Moreover, EBM outweighs GAMI-Net on both teachers, suggesting EBM are more capable of learning the teacher behaviour.

5.3 Supplemental metric - Error Fidelity

Teacher	Surrogate	Error Rate	Error Fidelity
Random Forest	EBM	0.0567	0.7673
Random Forest	GAMI-Net	0.0567	0.7367
Random Forest	Decision Tree Depth 5 (Baseline)	0.0567	0.7613
Random Forest	Logistic (Baseline)	0.0567	<u>0.6373</u>
Deep Neural Net	EBM	0.0851	0.7695
Deep Neural Net	GAMI-Net	0.0851	0.7575
Deep Neural Net	Decision Tree Depth 5 (Baseline)	0.0851	<u>0.6510</u>
Deep Neural Net	Logistic (Baseline)	0.0851	0.6803

Table 4: Error Fidelity comparison across surrogate models.

Although both EBM and GAMI-Net achieved high fidelity from table 2. Their error fidelity is at acceptance rate which about is about 75%. More specifically, EBM still outweighs GAMI-Net by a slight margin with 0.7673 to 0.7367 and 0.7695 to 0.7575. Both models also dominant the baseline of logistic regression.

5.4 Ablation Study Result

Teacher	Dataset	EBM ErrFid	GAMI-Net ErrFid
Random Forest	92k raw	0.7227	<u>0.7047</u>
Random Forest	300k perturbation	<u>0.7180</u>	0.7073
Random Forest	300k+VAE	0.7673	0.7367
Deep Neural Net	92k raw	<u>0.5115</u>	<u>0.6811</u>
Deep Neural Net	300k perturbation	0.7194	0.7682
Deep Neural Net	300k+VAE	0.7695	0.7575

Table 5: Ablation study on different data generation strategies.

We did ablation study on different data samples strategies where we remove part of our data samples use for training global surrogate models. From table 4, it can be observed that if only local perturbation is used, the error fidelity may be

hurt but mixing perturbation and VAE samplings, it can help improve the error fidelity.

6 Discussion

6.1 Fidelity may misleading

Lets look at an analogy. If two students are learning from a teacher, it does not guaranteed that they will have similar behaviour. For example, if 2 students are given a same test on different subject and both of them achieve very high score and answer the same on most of the question. It does not guaranteed that both of the students thinks and behave the same. For instance, student one can weak at Maths while the other weak at Biology. Our experiments from table reflects this analogy. Random Forest and Deep Neural Network teachers are trained on the same dataset with ground truth labels therefore although they have very high peer fidelity, their error peer fidelity is relatively low because they behave differently. This further suggest that if a model achieve very high accuracy, the pseudo-label produced by teacher model may looks identical to the ground-truth label. Therefore training global surrogate model on raw training samples with pseudo-labels may have the similar effect of training on ground-truth labels. Then, their fidelity may mis-leading without error fidelity. This emphasize the role of error fidelity to enforce the reliability of the surrogate model.

6.2 Ability of explaining and interpreting of surrogate models

With high fidelity, EBM and GAMI-NET agrees with original model most of the time, hence local interpretation derived from local sample is plausible most of the of the time. On the other hand, Error fidelity of both models are higher than peer error fidelity and baseline error fidelity, indicating that the surrogate models learns the behaviour of teacher model better than a linear relationship.

6.3 Limitations and Future Work

Although this experiments points out that fidelity may misleading and error fidelity can help enforcing the credibility of the surrogate models, single unified metrics are still missing that can reflect truly the credibility of the surrogate model. Furthermore, this experiment assume training data is accessible, but in some cases it is not. Therefore adaptive sampling methods can be used to query the original models for training data. In addition, in this experiment, we did not achieved high error fidelity and there is still a big gap between error fidelity and accuracy as well as F1 fidelity.

7 Conclusion

In this paper, we have experiment EBM and GAMI-Net as a global surrogate model to explain Random Forest and Deep Neural Network teacher models for

email phishing detection. In the experiment, we showed that both surrogate model are capable of explaining teacher models and we showed that error fidelity can enforce the reliability of the surrogate models. However, there are still a big gap between error fidelity and accuracy fidelity which future work can work on. Also, this may be a step forward to a single unified metric to measure the credibility of XAI methods.

Code availability

The source code and experimental implementation for this study are available at: <https://github.com/LuuMTran/Global-explainable-phishing-detection>

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A Additional Study: Qualitative results of surrogate models and discussion

A.1 Interpretation Plots

EBM vs GAMI-Net on Deep Neural Network Teacher

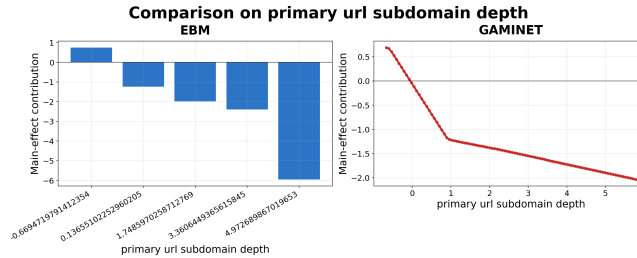


Fig. 5: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on primary subdomain depth

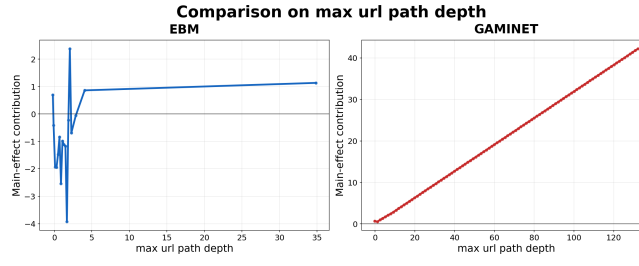


Fig. 6: Comparison of EBM (Left) and GAMINET (Right) interpretation on max url path depth

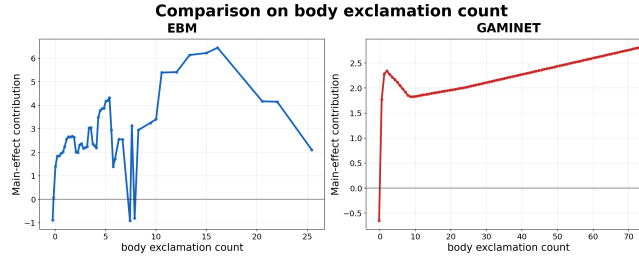


Fig. 7: Comparison of EBM (Left) and GAMINET (Right) interpretation on exclamation count

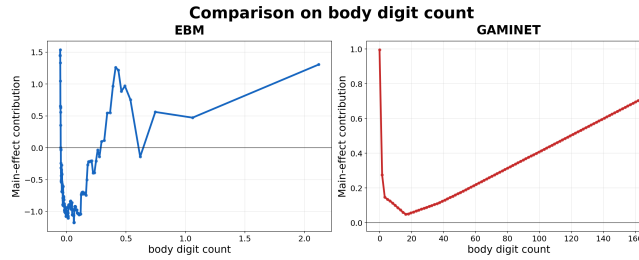


Fig. 8: Comparison of EBM (Left) and GAMINET (Right) interpretation on body digit count

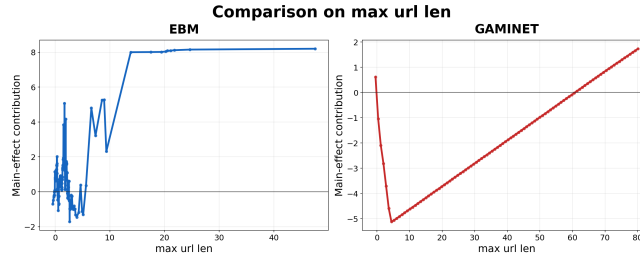


Fig. 9: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on max url length

The Figures 5 to 9 compares how the EBM and GAMI-Net explain the behaviours of the teacher model DNN by using different emails and URL features. These figures shows which features push the models labelling the email as phishing prediction and which features reduce the risk of being classified as phishing. EBM algorithm often provides a better, more detailed and threshold-based patterns, meanwhile GAMI-Net shows better smoothness in the trends. These figures help us to understand not just the result of if the email is phishing or not, but also the possible reasons behind the prediction by mimic the behaviour learned from the original black-box model.

Figure 5 shows that when the URL subdomain depth increases, both models reduce the phishing prediction. This suggests that the DNN teacher model treats deeper subdomain structures as a lower-risk feature in this testing dataset. The figure 6 shows a completely different pattern, as the URL path depth increases, both models suggest that the email is more likely to be phishing email. This is reasonable because that phishing emails often uses more complex URL links to make links look more legitimate. Figure 7 explains body exclamation count, which counts the number of exclamation marks in the email body. Both EBM and GAMI-Net suggest that as the exclamation mark count increases, the email is more likely to be classified as phishing which because phishing email often use urgent emotional language to pressure users into click into the links without thinking the content or notices any weird behaviour from the email. Figure 8 shows that boggy digit count has a very complex effect on the prediction, because both normal emails and phishing email can contain number, which might not be something can be use to predict the phishing email directly. Figure 9 shows that longer URLs strongly increase the prediction towards phishing. Especially in the EBM model, where the prediction becomes stronger after a certain threshold. Overall, these figures shows that EBM and GAMI-Net can explain how different features influence the DNN teacher model.

EBM vs GAMI-NET on Random Forest Teacher

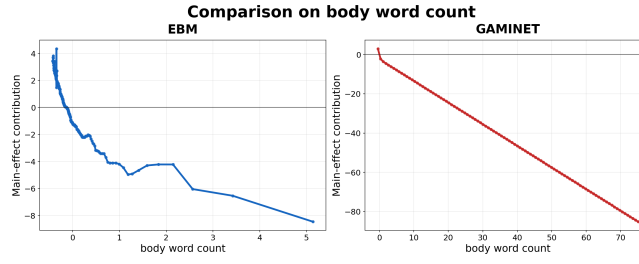


Fig. 10: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on body word count

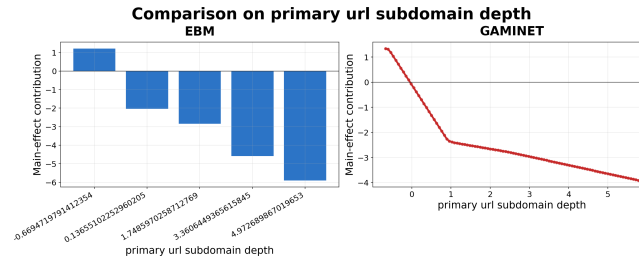


Fig. 11: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on url subdomain depth

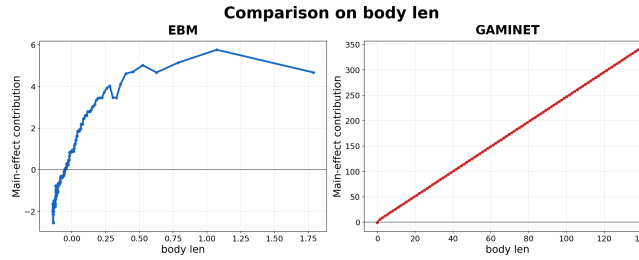


Fig. 12: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on body length

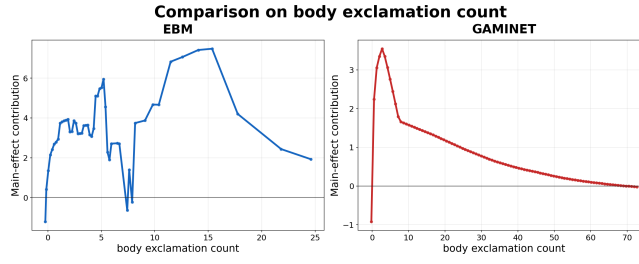


Fig. 13: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on exclamation count

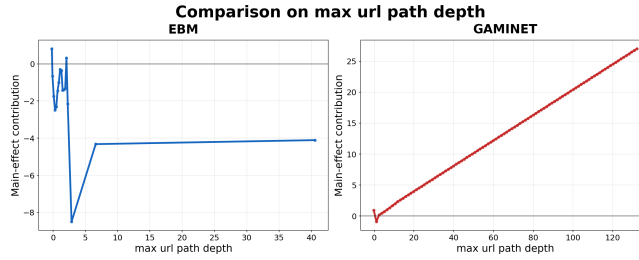


Fig. 14: Comparison of EBM (Left) and GAMI-NET (Right) interpretation on max url path depth

The interpretation plots in Figures 5 to 9 provide a direct comparison of how EBM and GAMI-Net approximate the behaviour of the Deep Neural Network (DNN) teacher model across the selected features. This section focuses on how each surrogate model represents those relationships.

Across all figures, a consistent difference can be observed in the structure of the learned feature contributions. EBM shows more segmented and step-like patterns, where the contribution changes more abruptly at certain feature values. In contrast, GAMI-Net produces smoother and more continuous curves, where the contribution changes gradually across the feature range. This distinction is visible in each of the plotted features.

This difference indicates that EBM represents the DNN teacher behaviour using localized value regions, where specific intervals of a feature correspond to distinct contribution levels. On the other hand, GAMI-Net represents the same behaviour using a continuous approximation, where the influence of a feature transitions smoothly rather than changing sharply.

Despite this structural difference, both models remain consistent in terms of overall directional behaviour across the figures. For each feature, EBM and GAMI-Net follow similar increasing, decreasing, or irregular patterns, indicating

that both surrogate models successfully capture how the DNN teacher model responds to changes in input features.

However, the way these patterns are expressed differs. EBM makes changes in contribution more explicit at specific ranges, while GAMI-Net reflects the same behaviour in a more gradual and smoothed form. This suggests that both models approximate the same underlying relationships, but through different functional representations.

Overall, the interpretation plots show that both EBM and GAMI-Net are able to approximate the DNN teacher model’s behaviour, while differing in how clearly feature contribution changes are represented.

B Additional Study: Comparison of explanation generation time compare to LIME and SHAP

Method	Mean (s)	Standard deviation(s)	Runtime for 100 samples
LIME on DNN	0.1399	0.0398	13.9906
LIME on Random Forest	0.2163	0.0335	21.6283
SHAP on DNN	0.0216	0.0036	2.1584
SHAP on Random Forest	0.2790	0.0334	278.9888
EBM	0.0137	0.0017	1.3711
GAMI-NET	0.0109	0.0567	1.094

Table 6: Comparison of different method’s explanation time

This additional study compares the interpretation latency of SHAP and LIME with the proposed surrogate explanation methods, EBM and GAMI-Net, to evaluate whether global surrogate models can provide faster explanations for phishing email detection while maintaining interpretability. The results show that both SHAP and LIME required substantially more time to generate explanations, especially when applied repeatedly to the DNN and Random Forest teacher models with 292 input features. SHAP was particularly expensive for the Random Forest teacher. In contrast, EBM and GAMI-Net generated explanations much faster because most of the computational cost was shifted to the offline surrogate-training stage, allowing deployment-time explanations to be produced directly from the trained additive surrogate structures. Although SHAP and LIME remain useful for detailed local post-hoc interpretation, the findings suggest that they may become a practical bottleneck when many phishing emails or analyst-selected cases need to be explained. Therefore, the latency comparison strengthens the main argument of this research by showing that faithful global surrogate models can provide a more scalable explanation mechanism, offering strong interpretability and much lower explanation cost, as long as their latency advantage is interpreted together with fidelity and error-fidelity results.